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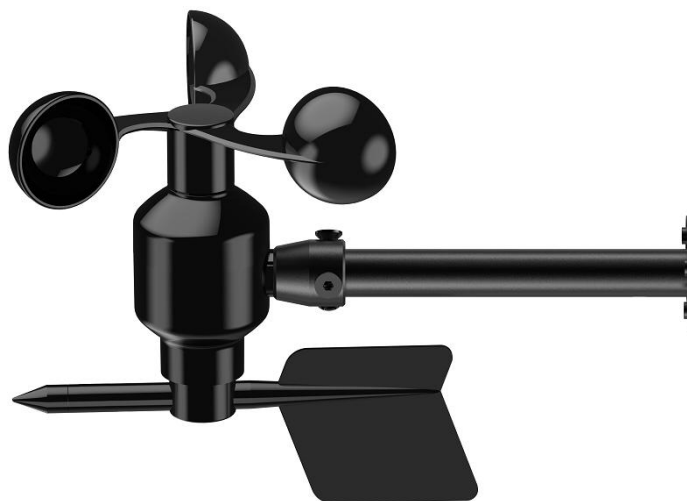
XS-WSDS01 Small Integrated Wind Speed and Direction Sensor (Polycarbonate)V1.2

Product Manual

Mianyang Xiangshu Electronic Technology Co., LTD



1. Product Introduction



Integrated wind speed and direction sensor is a wind speed and direction measuring instrument independently developed and produced by our company. The sensor shell is made of polycarbonate composite material, which has good anti-corrosion characteristics, which can ensure long-term use of the sensor without rust. At the same time, it cooperates with the internal smooth bearing system to ensure the stability of information collection. Internal integration of photoelectric conversion mechanism, industrial microcomputer processor, standard current generator, current driver, etc.

The circuit is designed according to EMI specifications, making the whole with extremely reliable anti-electromagnetic interference ability, and the electronic components are imported industrial-grade chips to ensure the stability of the measurement parameters and electrical performance.

Features:

- (1) Compact design with high measurement accuracy.
- (2) Equipped with an optional heating function, it is not afraid of low temperatures and can still operate normally at -40°C.
- (3) Truly achieve low cost, low price and high performance.
- (4) Wide supply voltage range of 5 to 24V, RS485 is the standard Modbus-RTU protocol.
- (5) Wind speed parameters: wind speed (m/s), wind scale, wind speed (km/h).
- (6) Wind direction parameters: Wind direction Angle, wind direction direction (8-direction or 16-direction).

Application:

This product is widely used in construction machinery, railway, port, wharf, power plant, meteorology, ropeway, environment, greenhouse, breeding and other fields for measuring wind speed and wind direction.

2. Technical Parameters

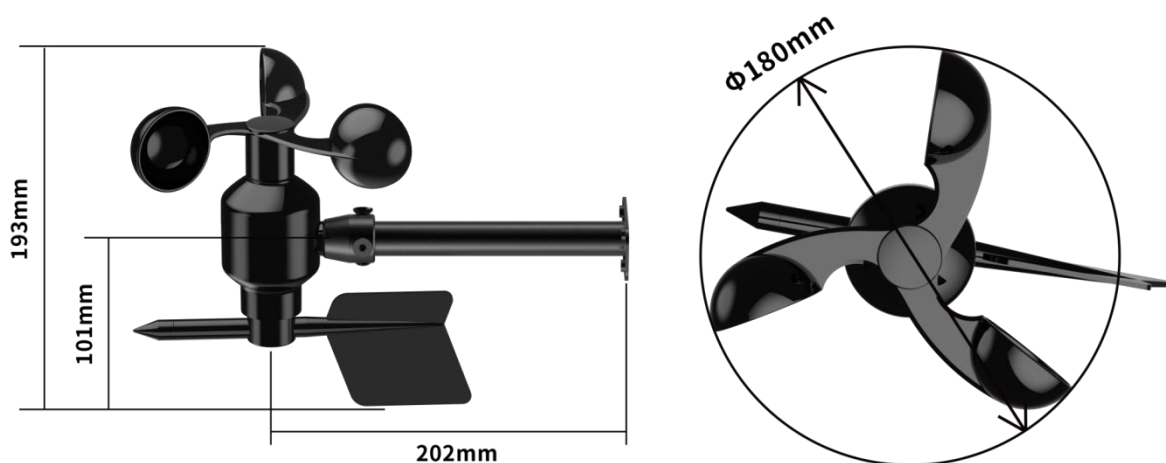
Parameter	Specification / Description
Wind Speed Parameters	Wind speed (m/s), Wind scale, Wind speed (km/h)
Wind Direction Parameters	Wind direction angle, Wind direction orientation (8 or 16 positions)
Wind Speed Range	0~30 m/s (default) (customizable up to 60 m/s)
Wind Direction Range	0~360°
Starting Wind Speed	≤0.3 m/s
Accuracy	±(0.3+0.03V) m/s; ±1°
Resolution	0.1 m/s; 0.1°
Product Weight	0.52 kg (with 1-meter cable and bracket)
Housing Material	Polycarbonate composite
Bearing Material	304 Stainless Steel
Output Signal	A: RS485 (standard Modbus-RTU protocol, default device address: 01)
	B: 4~20mA (current loop)
	C: Voltage signal (0~2V; 0~5V; 0~10V)
Power Supply Voltage	5~24V DC (when output signal is RS485, 0~2V, and no heating)
	12~24V DC (when output signal is 4~20mA, 0~5V, or 0~10V)
	12~24V DC (when heating function is enabled)
Operating Current	No heating version: 23mA @ 5V;
	With heating version: 165mA @ 12V; 325mA @ 24V
Protection Measures	1. Reverse connection protection for power supply 2. Electrostatic Discharge Immunity: Contact discharge: GB/T17626.2-2018, Level 1, 2kV Air discharge: GB/T17626.2-2018, Level 2, 4kV 3. Electrical Fast Transient/Burst Immunity: Signal port: GB/T17626.4-2018, I/O port Level 2, 0.5kV (5/50ns, 5kHz) 4. Surge Immunity: GB/T17626.5-2019, Level 1, 0.5kV (line to ground, 1.2/50μs, 8/20μs combined wave)
Installation Method	Bracket flange mounting (three types of brackets)
Operating Temperature	0~70°C (no heating version); -40~70°C (with heating version)
Storage Temperature	-40~70°C
Waterproof Rating	IP65
Stabilization Time	<1s

Response Time	<1s
Cable Length	1m (AVVR4*0.2) (can be extended as needed)

Impedance Requirement of Current Signal

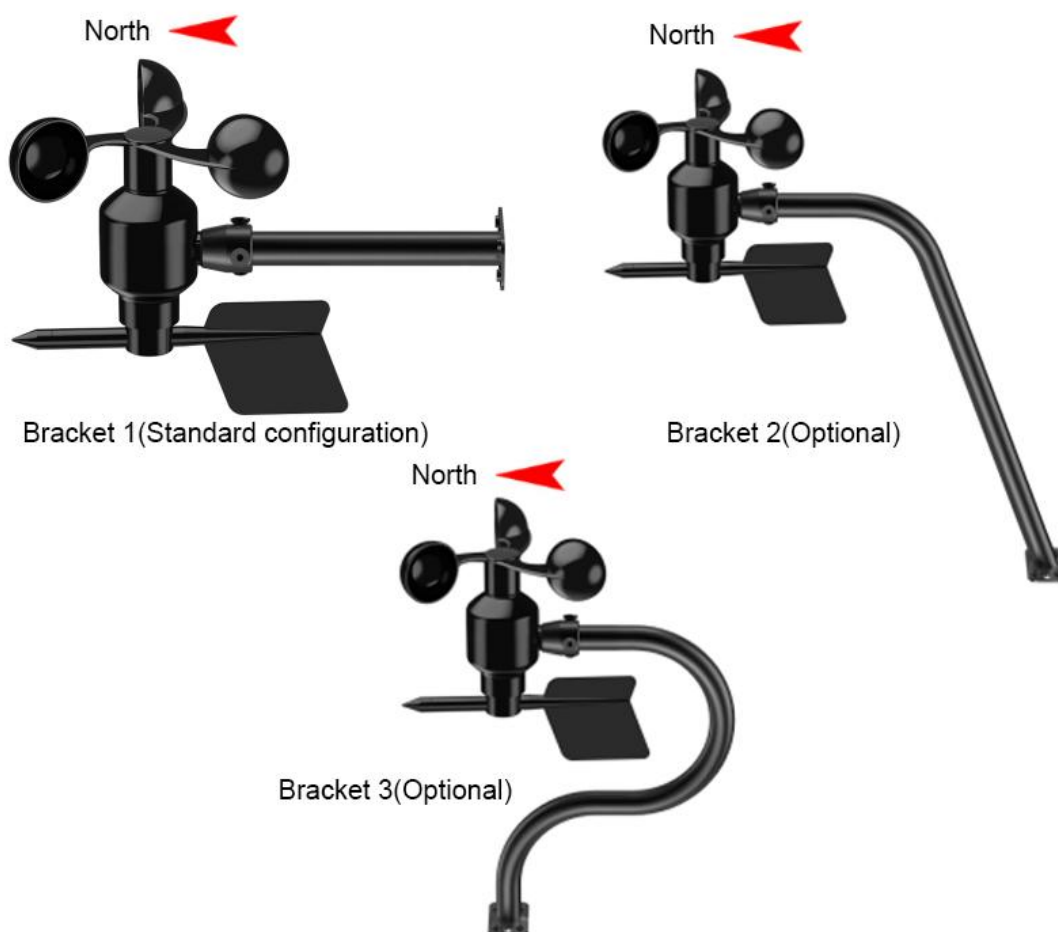
Impedance requirements for current (4 ~ 20mA) signals				
Power Supply Voltage	9V	12V	18V	24V
Maximum Impedance	150Ω	300Ω	600Ω	900Ω

3. Dimension



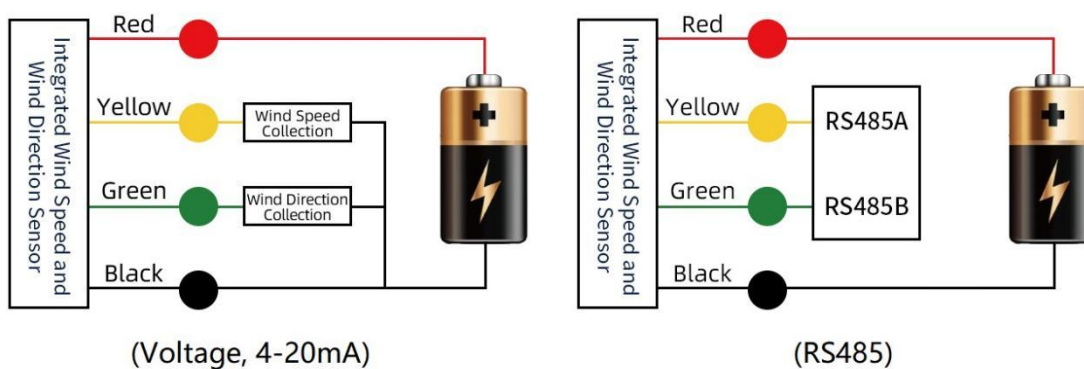
4. Installation

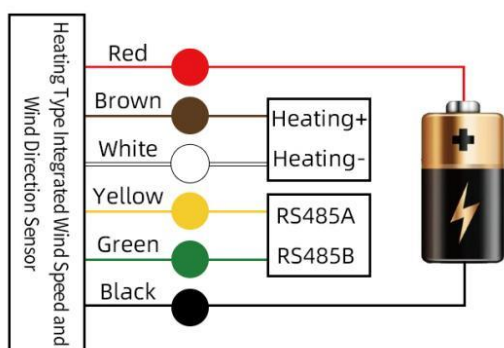
Integrated Wind Speed and Wind Direction Sensor (wire outlet points south, the opposite direction is north), the installation method is as below:



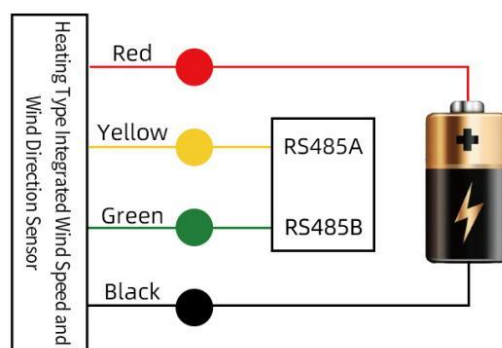
5. Wiring method

Integrated Wind Speed and Wind Direction Sensor can be connected to various data collectors with differential inputs, data acquisition cards, remote data acquisition modules and other equipment. The wiring instructions are as follows:

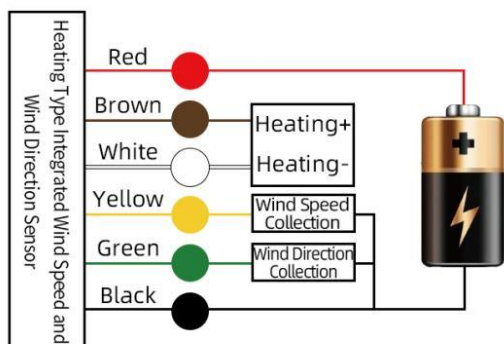




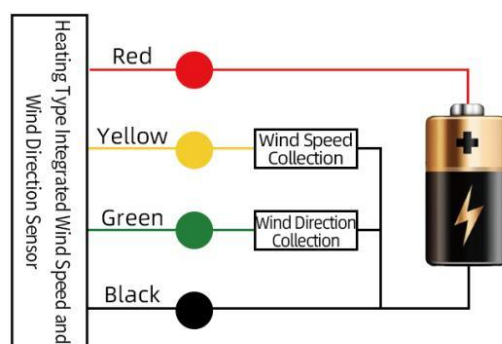
(RS485 Two Channels Power Supply)



(RS485 Single Channel Power Supply)



(Voltage、4-20mA Two Channels Power Supply)



(Voltage、4-20mA Single Channel Power Supply)

6. Data Conversion Method

V: voltage value collected by the collector, unit: V;

A: The current value collected by the collector, unit: mA;

Output Signal	Data Conversion Method for Each Range of Wind Speed	
	0~30m/s	0~60m/s
0~2V DC	Wind speed=15*V	Wind speed=30*V
0~5V DC	Wind speed=6*V	Wind speed=12*V
0~10V DC	Wind speed=3*V	Wind speed=6*V
4~20mA	Wind speed=1.875*A-7.5	Wind speed=3.75*A-15

Data Conversion Method for Each Range of Wind Direction					
	0~2V	0~5V	0~10V	4~20mA	RS485
North	1.938~0.063	4.844~0.156	9.688~0.313	19.5~4.5	0X00
North Northeast	0.063~0.188	0.156~0.469	0.313~0.938	4.5~5.5	0X01
Northeast	0.188~0.313	0.469~0.781	0.938~1.563	5.5~6.5	0X02
Northeast China	0.313~0.438	0.781~1.094	1.563~2.188	6.5~7.5	0X03

East	0.438~0.563	1.094~1.406	2.188~2.813	7.5~8.5	0X04
Southeast Of The East	0.563~0.688	1.406~1.719	2.813~3.438	8.5~9.5	0X05
Southeast	0.688~0.813	1.719~2.031	3.438~4.063	9.5~10.5	0X06
South, East, South	0.813~0.938	2.031~2.344	4.063~4.688	10.5~11.5	0X07
South	0.938~1.063	2.344~2.656	4.688~5.313	11.5~12.5	0X08
South Southwest	1.063~1.188	2.656~2.969	5.313~5.938	12.5~13.5	0X09
Southwest	1.188~1.313	2.969~3.281	5.938~6.563	13.5~14.5	0X0A
Southwest	1.313~1.438	3.281~3.594	6.563~7.188	14.5~15.5	0X0B
West	1.438~1.563	3.594~3.906	7.188~7.813	15.5~16.5	0X0C
Northwest Northwest	1.563~1.688	3.906~4.219	7.813~8.438	16.5~17.5	0X0D
Northwest	1.688~1.813	4.219~4.531	8.438~9.063	17.5~18.5	0X0E
North Northwest	1.813~1.938	4.531~4.844	9.063~9.688	18.5~19.5	0X0F

RS485 signal (Default address 01):

Standard Modbus-RTU protocol, baud rate: 9600; check bit: no; data bit: 8; stop bit: 1

PLC Configur ation Address	Modbus Register Address	Register Description	Data Type 16-bit Integer	Data Description (Signed integer if negative values exist)	Default Value
40001	0x0000	Wind Speed (m/s)	Read-only	1 decimal place, 0–30 m/s	None
40002	0x0001	Wind Level	Read-only	No decimal places	None
40003	0x0002	Wind Speed (km/h)	Read-only	2 decimal place, 0–108 km/h	None
40004	0x0003	Wind Direction Angle	Read-only	1 decimal place, 0–360°	None
40005	0x0004	Wind Direction Bearing	Read-only	8 or 16 bearings	None
40049	0x0030	Device Address	Read-Write	1–253	1
40050	0x0031	Baud Rate	Read-Write	5 types	9600
40051	0x0032	Parity Bit	Read-Write	0: None / 1: Odd / 2: Even	0: No parity
40052	0x0033	Auto Report	Read-Write	0–65535 seconds	0: No reporting
40108	0x006B	Heating Function Enable	Read-Write	0: Off; 1: On	Heating off

40109	0x006C	Heating Start Temperature	Read-Write	1 decimal place, negative values in two's complement	5°C
40110	0x006D	Heating Stop Temperature	Read-Write	1 decimal place, negative values in two's complement	25°C
40111	0X006E	Calibrate with the wind direction pointing north	Read-Write	0X0001	None

6.1 Modification Address

For example: change the address of the sensor with address 1 to 2, master→slave

Original Address	Function Code	Start Register High	Start Register Low	Start Address High	Start Address Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X30	0X00	0X02	0X08	0X04

If the data returned from the slave machine to the host is the same as the sent data, it indicates that the Settings were successful.

Note: If the original address of the sensor is forgotten, the broadcast address 0XFE can be used as a substitute. When using the broadcast address 0XFE, the host can only be connected to one slave.

6.2 Query Data

If the sensor address is forgotten, a query address instruction can be sent using the broadcast address 0XFE, with the host → slave

Broadcast address	Function Code	Start Register Address High	Start Register Address Low	Address Data High	Address Data Low	CRC16 Low	CRC16 High
0XFE	0X03	0X00	0X30	0X00	0X01	0X90	0X0A

If the sensor receives correctly, the following data will be returned, slave→host.

Broadcast address	Function Code	Data Length	Register 0 Data High	Register 0 Data Low	CRC16 Low	CRC16 High
0XFE	0X03	0X02	0X00	0X02	0X2D	0X91
Sensor address: 02						

6.3 Modify the baud rate

For example: Set the baud rate of the sensor at address 1 to 4800, master → slave

Address	Function Code	Starting Register High	Starting Register Low	Baud Rate Data High	Baud Rate Data Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X31	0X48	0X00	0XEE	0X05

				0X4800: Indicates baud rate 4800 0X9600: Indicates baud rate 9600 (default baud rate is 9600) 0X1920: Indicates baud rate 19200 0X5760: Indicates baud rate 57600 0X1152: Indicates baud rate 115200		
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If the data returned from the slave machine to the host is the same as the sent data, it indicates that the Settings were successful.

6.4 Query Data

Query Sensor (Address 1) Data (5 Parameters for Wind Speed and Direction), Host → Slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	Register Length High	Register Length Low	CRC16 Low	CRC16 High
0X01	0X03	0X00	0X00	0X00	0X05	0X85	0XC9

If the sensor receives it correctly, it returns the following data, slave → host

Address	0X01	
Function Code	0X03	
Data Length	0X0A	
Register 0 Data High	0X00	Wind Speed: 3.8 m/s
Register 0 Data Low	0X26	
Register 1 Data High	0X00	Wind Level: Level 3
Register 1 Data Low	0X03	
Register 2 Data High	0X05	Wind Speed: 13.68 km/h
Register 2 Data Low	0X58	
Register 3 Data High	0X01	Wind Direction Angle: 45.0 °
Register 3 Data Low	0XC2	
Register 4 Data High	0X00	Wind Direction Orientation: Northeast
Register 4 Data Low	0X02	
CRC16 Low	0XA5	
CRC16 High	0X8E	

6.5 Set Active Reporting

Example: Set the sensor with address 1 to report data every 2 seconds, host → slave

Address	Function Code	Starting Register High	Starting Register Low	Active Reporting Time Data High	Active Reporting Time Data Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X33	0X00	0X02	0XF8	0X04
				0X0002: Report every 2s 0X0000: Indicates active reporting is off (default is off)			

If the data returned from the slave to the host matches the sent data, the setting is successful.

6.6 Query Automatic Heating Temperature Threshold

Query the heating temperature threshold of the sensor (address 1), host → slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	Register Length High	Register Length Low	CRC16 Low	CRC16 High
0X01	0X03	0X00	0X6C	0X00	0X02	0X04	0X16

If the sensor receives it correctly, it returns the following data, slave → host

Address	Function Code	Data Length	Register 0 Data High	Register 0 Data Low	Register 1 Data High	Register 1 Data Low	CRC16 Low	CRC16 High
0X01	0X03	0X04	0X01	0X2C	0X00	0X32	0XBB	0XD3
			Stop Heating Temperature: 30°C		Start Heating Temperature: 5°C			

Note: The default setting is to start heating when the temperature $\leq 5^{\circ}\text{C}$ and stop heating when the temperature $\geq 30^{\circ}\text{C}$.

To modify the temperature thresholds, refer to the following protocol.

6.7 Set Automatic Heating Temperature Threshold

6.7.1 Set Start Heating Temperature Threshold

Set the start heating temperature threshold for the sensor (address 1), host → slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	Start Temperature Data High	Start Temperature Data Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X6C	0X00	0X0A	0XC9	0XD0
				Start Heating Temperature: 1.0°C When the temperature $\leq 1.0^{\circ}\text{C}$, start heating Note: Negative temperatures are represented in two's complement.			

If the data returned from the slave to the host matches the sent data, the setting is successful.

6.7.2 Set Stop Heating Temperature Threshold

Set the stop heating temperature threshold for the sensor (address 1), host → slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	Stop Temperature Data High	Stop Temperature Data Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X6D	0X01	0X18	0X19	0X8D
				Stop Heating Temperature: 28.0°C When the temperature ≥ 28.0°C, stop heating			

If the data returned from the slave to the host matches the sent data, the setting is successful.

Note: The start temperature must be less than the stop temperature; otherwise, the heating function will not activate.

6.8 Set Automatic Heating Function Enable

Set the automatic heating function enable for the sensor (address 1), host → slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	Automatic Heating Function Enable Data High	Automatic Heating Function Enable Data Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X6B	0X00	0X01	0X39	0XD6
				0X0000: Disable automatic heating function 0X0001: Enable automatic heating function			

If the data returned from the slave to the host matches the sent data, the setting is successful.

Note: The automatic heating function enable acts as a heating switch.

It only activates the heating function when set to enable and the temperature is below the threshold; otherwise, the heating does not start.

6.9 Query Automatic Heating Function Enable

Query the automatic heating function enable of the sensor (address 1), host → slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	Register Length High	Register Length Low	CRC16 Low	CRC16 High
0X01	0X03	0X00	0X6B	0X00	0X01	0XF5	0XD6

If the sensor receives it correctly, it returns the following data, slave → host

Address	Function Code	Data Length	Register 0 Data High	Register 0 Data Low	CRC16 Low	CRC16 High
0X01	0X03	0X02	0X00	0X01	0X79	0X84
			0X0000: Disable automatic heating function 0X0001: Enable automatic heating function			

6.10 Wind Direction North Setting

If the sensor (address 1) cannot be installed with the north marker pointing north during installation, you can reset the north orientation by pointing the wind vane arrow north and sending the north setting instruction,

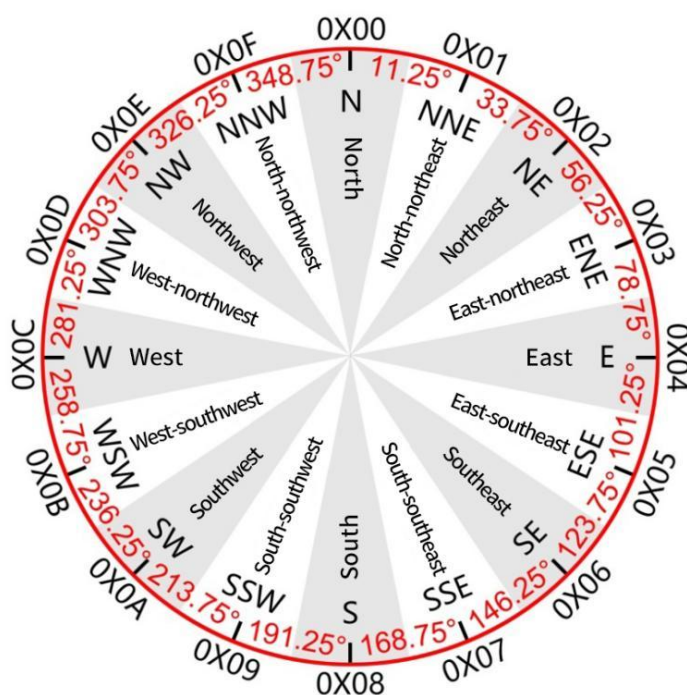
host → slave

Address	Function Code	Starting Register Address High	Starting Register Address Low	North Data High	North Data Low	CRC16 Low	CRC16 High
0X01	0X06	0X00	0X6E	0X00	0X01	0X29	0XD7

If the data returned from the slave to the host matches the sent data, the setting is successful.

7.Wind Direction 16 Azimuth Map

16 Orientation Diagram of Wind Direction Sensor



Wind Scale	Name of Wind	Wind Speed (m/s)	(km/h)	Land Phenomenon	Sea State
0	No wind	0-0.2	<1	Quiet, smoke straight up	Calm as a mirror
1	Light air	0.3-1.5	1-5	The smoke can indicate the direction of the wind, but the wind vane cannot turn.	Tiny waves
2	Light air	1.6-3.3	6-11	The human face feels windy, the leaves make a slight noise, and the weather Vane can turn.	Wavelet
3	Breeze	3.4-5.4	12-19	The leaves and twigs swayed endlessly, and the flag unfolded.	Wavelet
4	Soft breeze	5.5-7.9	20-28	Can blow up dust and paper on the ground, and the twigs of the tree move slightly.	Slight sea
5	Fresh breeze	8.0-10.7	29-38	The leafy twigs sway, and the inland water has wavelets.	Moderate sea
6	Strong breeze	10.8-13.8	39-49	Big branches swing, electric wires whirring, difficult to lift umbrellas.	Rough sea
7	Strong wind	13.9-17.1	50-61	The whole tree is swaying, it is inconvenient to walk in the wind.	Surge
8	Gale	17.2-20.7	62-74	The twigs are broken, and people feel a lot of resistance to move forward.	Fierce waves
9	Strong gale	20.8-24.4	75-88	The building is damaged (the top of the chimney and the roof tiles move).	Violent waves
10	Whole gale	24.5-28.4	89-102	Rare on land, trees can be uprooted when seen, causing serious damage to buildings.	Violent waves
11	Storm wind	28.5-32.6	103-117	Rare on land, trees can be uprooted when seen, causing serious damage to buildings.	Special Phenomenon
12	Hurricane	32.7-36.9	118-133	Very few on land, its destructive power is extremely high.	Special Phenomenon
13	Hurricane	37.0-41.4	134-149	Very few on land, its destructive power is extremely high.	Special Phenomenon
14	Hurricane	41.5-46.1	150-166	Very few on land, its destructive power is extremely high.	Special Phenomenon
15	Hurricane	46.2-50.9	167-183	Very few on land, its destructive power is extremely high.	Special Phenomenon
16	Hurricane	51.0-56.0	184-201	Very few on land, its destructive power is extremely high.	Special Phenomenon
17	Hurricane	56.1-61.2	202-220	Very few on land, its destructive power is extremely high.	Special Phenomenon

9. Precautions

- Please check whether the packaging is in good condition, and check whether the sensor model and specifications are consistent with the products you purchased.
- No live wiring is allowed, and the power can be turned on after the wiring is checked and checked.
- The user should not change the components and wires that have been soldered after the product leaves the factory.
- The sensor is a precision device, please do not disassemble it by yourself when using it, let alone touch the diaphragm, so as to avoid damage to the product.
- Avoid sticky particles from entering the inside of the sensor and protect it from humidity, so as not to affect the measurement performance.

Thank you for purchasing our products!

Due to the continuous improvement of our products, the products you purchase may differ from the illustrations in the manual without notice. Please refer to the actual product.

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Contact us



Contact us

Units: Mianyang Xiangshu Electronic Technology Co., LTD
Sichuan Veinasa Technology Co., Ltd.

Contact: Mr. Tang

Phone: 86-0816-2712180

Fax: 86-0816-2712180

E-mail: info@veinasa.com

WhatsApp/Skype/Wechat: 86-18161001208

Address: Innovation Center, Mianyang Science and Technology City New District,
Mianyang City, Sichuan Province